

Nullius

nullius in verba

“Take nobody’s word for it.”

The Royal Society’s motto since 1660, and the oldest continuous statement of the discipline this harness enforces. Then the unreliable narrator was authority. Now it is a fluent model, an author’s own abstract, and you. Same job.

This document is the argument, not the manual. It says what an AI assistant does wrong in research work by default, why writing better instructions does not fix it, and what a harness has to constrain instead.

The short version: research has no compiler, so the design problem is finding what plays that part. Seven things do. Everything else in the tool is built on top of them, and nothing that cannot fail is allowed to block.

1 The motto is the specification

Names are usually labels. This one is a specification. *Nullius in verba* was adopted by the Royal Society in 1660 as a refusal to settle questions by citing authority, and every constraint in this harness turns out to be that sentence pointed at a different narrator.

WHY IT IS NOT DECORATION
Every mechanism in the design is an instance of one sentence. That is a rare property in a name, and it doubles as a consistency check: a mechanism that is not a conjugation of the motto probably does not belong.

DO NOT TAKE	THE MECHANISM
the author’s word	an authors–claim may never be written as a bare assertion
the abstract’s word	warrant is capped by how deeply you actually read
the field’s word	established is computed from author-set overlap, never declared
the repeated word	folklore is a citation trail that never reaches data
your own word	your unpublished data is held to a published paper’s bar
the reviewer’s word	a defensible finding closes with one sentence, permanently

The last row is the one people are surprised by, and it is the reason this is a harness rather than a critic. Taking nobody’s word for it cuts both ways:

an unexamined objection is no better than an unexamined claim, and a tool that can always find one more thing wrong has stopped providing evidence and started providing pressure.

2 What goes wrong by default

None of this is the model being lazy. Each is a place where nothing in the loop *could* fail, so the default filled the gap.

It skims the literature and does not know it

Four papers, one query, one afternoon, written up as “*a review of the literature on X*”. The prose form hides it perfectly: one query and forty look identical on the page. Worse, the search runs in *your* vocabulary rather than the field’s, so an entire literature can sit one synonym away and never appear.

It trusts a paper the way the paper describes itself

An abstract is a marketing document. What a study *shows* is in its tables, and the gap between the two is where most of the interesting reading happens. By default that gap is collapsed: the claimed contribution and the evidenced contribution arrive as one sentence.

It cannot tell settled from proposed

This is the expensive one. “*Attention improves long-range modelling*” and “*method Y improves long-range modelling*” come out in the same declarative mood at the same confidence. One carries a decade of independent replication; the other carries one paper and one benchmark. Build three chapters on the second believing it was the first, and you find out late.

Its critique has no floor

Asked what is missing, it answers. Asked again, it answers again. *More* reads as *more rigorous*, so a finished draft becomes an infinite project, and a page limit that was never declared cannot object. The 12-to-24 story above is not a pathology; it is the predictable output of an unbounded question asked four times.

OBSERVED, NOT IMAGINED

These are failures seen in real sessions. The clearest one: a research proposal that grew from 12 pages to 24 across three runs, where the fourth still returned a list of what was missing.

THE ASYMMETRY THAT MAKES THIS HARD

Suppress “*what is missing*” and a genuinely absent control section goes unmentioned. Leave it unbounded and the draft doubles and

is still called incomplete. A single rule cannot serve both, which is why the two have to stop being one axis.

3 Why more instructions do not fix it

The instinct is to write the rule more firmly. *Search thoroughly. Verify before claiming. Do not overstate.* These are not wrong. They constrain the wrong thing.

A rule that describes *how* to work has no truth value. Nothing can check it, so it competes for attention with everything else in the context and loses whenever the task gets interesting. A constraint on *what must hold* has a truth value: it can be observed, it costs nothing until it fires, and it leaves the approach entirely alone.

MEASURED, NOT ASSERTED

The predecessor of this design replaced a 43,494-word instruction collection, plus 9,572 words loaded on every turn, with six hooks and one ledger. Every rule in the prose was a request; none could fail. That is why a rule had to be repeated three to five times before it took effect.

PROCESS CONSTRAINT · UNENFORCEABLE	OUTCOME CONSTRAINT · ENFORCEABLE
"survey the literature first"	a survey unit cannot close with unscreened > 0
"don't over-trust the paper"	a claim may not exceed its source's read depth
"distinguish settled from new"	established requires disjoint author sets
"cite accurately"	an unresolved identifier cannot reach the draft
"say what done means"	an acceptance question that is open right now
"don't pad it"	over budget, an addition must name what it costs
"know when to stop"	zero material findings is a verdict, not a shorter list

Each row on the right is a state the session can be observed to be in. That is the whole difference, and it is why this is a set of gates rather than a longer prompt.

4 Research has no compiler

Software gets this for free. *It builds, the suite is green, the check went red to green* are facts, observed rather than reported, and facts can block. Research has no equivalent, so the entire design question is: which questions here are *equally* objective?

Seven are, and every one is a lookup or a count. None asks the model to judge quality.

THE HONEST FLOOR

It is a short list, and it is enough. Shallow is invisible in prose and undeniable in a row of numbers.

QUESTION	THE CHECK
Does this reference exist?	the identifier resolves, or it does not
Was it retracted or corrected?	retraction metadata on the record
Does the paper say this?	the quote is verbatim in the cached text, or it is not
Are these two sources independent?	author and institution sets: disjoint, or not
Does this claim's trail reach data?	walk the references back and see where it lands
Did you search the field's words?	distinct query vocabularies, with hit counts
Is a required section absent?	the venue's own list, walked

Two of those are the cheapest real epistemics available and nobody builds them. **Source independence is set arithmetic on author lists. Folklore is a citation trail that never lands on evidence.** Both fall out of metadata already fetched to render a bibliography.

```
$ nullius coverage
found      231   screened 0   included  4
vocabularies 1   of 3 required --- try the field's terms, not yours
groups     1   4 works, 2 shared authors, not independent
years      2024—2026  most-cited ancestor of the seed set unread
```

“I reviewed the literature” survives any amount of scrutiny. That block does not.

5 Warrant, capped by read depth

Every claim carries a warrant class saying what settles it. The class that matters most is the one nobody tracks: authors-claim, meaning *the authors assert this*, recorded separately from what their tables show.

THE STRONGEST SINGLE RULE
You cannot talk your way past it, because the note recorded how far you read *before* the claim existed.

Then the cap. A paper note records how far you actually got — abstract, skim, method, replicated — and **a claim may not exceed the read depth of its source.**

READ DEPTH	“THEY REPORT X”	“X HOLDS”	“THEIR METHOD SHOWS WHY”	“THIS GENERALISES”
abstract	ok	refused	refused	refused
skim	ok	ok	refused	refused
method	ok	ok	ok	refused
replicated	ok	ok	ok	ok

This works where an instruction does not because the difference is *textual*. *“They report a 12% gain”* and *“the method gives a 12% gain”* are different strings, and the ledger holds the class. A script can decide it. *“Be appropriately skeptical”* cannot be decided by anything.

6 Settled versus proposed

Epistemic status is a second axis, and it is the one that costs years when it is missing. The failure is not citing something false; it is citing something *true but unsettled* as though it were settled.

ORTHOGONAL AXES

Warrant asks *how do you know this*.
Status asks *how settled is this in the field*.

STATUS	WHAT EARNS IT	HOW IT MAY APPEAR IN PROSE
textbook	graduate texts treat it as background	bare assertion
established	<i>independent</i> replication, disjoint author sets	bare assertion, review cited
contested	serious work on both sides, both logged	must name both sides
emerging	one or two groups, recent	"early evidence suggests", groups counted
single-result	one paper, one setting	attributed and situated, never bare
folklore	repeated; the trail never reaches data	marked unsourced, or dropped

Two things stop status from being self-declared. It is **capped by source independence** — the exact parallel to the read-depth cap, and computable from author identifiers the record already carries. And its rendering is greppable, so the discipline lives in a difference a script can see rather than in an adjective the model is asked to feel.

Pointed the other way, the same axis is what stops your open question from being closed on your behalf: a live disagreement rendered as contested keeps a door open that a confident summary would have quietly shut.

7 Missing, or merely more

Back to the asymmetry. The resolution is that "*something is missing*" is three different claims wearing one word, and they differ in whether they can point at something enumerable.

THE HARDEST ARBITRATION

And the one the 12-to-24 story needed — note that it is not a judgment call.

KIND	GROUNDING IN	WHY IT TERMINATES
structural	the venue's list of required sections	the checklist is finite
evidential	the claim ledger	the claims are enumerable
coherence	two locations in the artifact	the pairs are finite
enhancement	nothing	it never does

So: a finding must cite a required-section id, a claim id with an empty warrant, or two conflicting locations. A finding that cites none of the three is enhancement, and inadmissible. No severity intuition required — a finding either has a referent or it does not.

The same rule keeps the tool honest in the other direction. The complete-

ness sweep is **walked, not sampled**: every required section gets present, thin or absent, every run. “*It never told me the timeline was missing*” cannot happen, because the checklist is enumerated rather than intuited. And “required” comes from a real document — the grant call, the author guidelines, the venue’s checklist, the field’s reporting standard — so the tool cannot invent a requirement it can’t cite.

The budget is a price, not a ceiling

Now the two axes compose. A gap is real or it is not; the artifact is inside its format or over it. Four situations, one honest answer each.

	INSIDE THE BUDGET	OVER THE BUDGET
A REAL GAP EXISTS	Say it. Additions admissible. The gap is real and there is room. The ordinary case, and nothing here suppresses it.	Say it – and name what it costs. Both are true at once, so the fix is a <i>trade</i>. “The sampling frame is unstated; the two paragraphs on X are the cheapest 150 words.”
NO REAL GAP	submit Zero structural, evidential and coherence gaps, inside the format. A verdict – and the tool is forbidden from producing further findings.	Only cuts. Nothing is missing and it is too long. Now, and only now, the question becomes what earns its place.

The upper-right cell is the one that fourth run needed and did not have. And the funding requirement is a severity filter the tool **cannot fake**: nobody trades two working paragraphs for a nice-to-have, so soft findings collapse under their own price without anyone having to judge them soft.

THE REFRAME

The budget does not forbid additions. It **prices** them — and a priced list has to be ranked, while a free list can only be enumerated. An unbounded set of improvements becomes a ranked set of trades, and a ranked set has a top and therefore a bottom.

The switch from “*what is missing*” to “*what earns its place*” is then nobody’s judgement call. It happens on its own, the moment the last real gap closes.

8 What blocks, and what is merely reported

A fact and a chosen number are different claims, and conflating them is how a good gate turns into a nuisance. Facts block. Thresholds that were picked rather than measured are reported, and marked as chosen.

WHY THE SPLIT MATTERS MORE HERE

Almost everything worth saying about research is a calibrated judgement. If those blocked, the harness would become the thing it exists to prevent.

FACTS · THESE REFUSE THE STOP

acceptance question still open
a citation unresolved, or retracted
a quote that is not verbatim
a claim above its source's read depth
a status stronger than independence earns
a required section never walked
over the format budget with no trade named
a critique with no recommendation

CHOSEN · THESE ARE REPORTED

a citation trail that never reaches data
source independence looking thin
passes with no new external object
a one-sided support/refute balance
the extension-loop signal
single-source claims present
judge scores, for ranking only

The left column needs no calibration to be right. The right column would need a hundred or two labelled examples and an agreement measure before it could honestly refuse anything, and that work has not been done — so it does not refuse. Saying which is which is not a caveat. It is the difference between a gate you trust and one you learn to route around.

9 Making it yours

A harness that hard-coded one field's norms would be useless in every other field and quietly wrong in its own. So the tool ships knowing nothing, and the norms are configuration:

- `field.md` — your subfield's norms. What a normal sample size is here, what a standard baseline is, which claims need what kind of evidence. This is what makes a critique *calibrated* rather than idealised.
- `venues/<venue>.md` — required sections, page limit, what reviewers there actually ask for. Bootstrapped once from the real call or author guidelines. A structural finding must cite a line in this file, so the tool cannot invent a requirement.
- `program.md` — what you are actually working on, so a thread that does not trace to it can be flagged as the distraction it is.
- `threads/, papers/, claims.md, falsified.md` — the durable record. This is the half that survives a dead context window, and a dead semester.

THE ONE FILE PEOPLE UNDERESTIMATE

`falsified.md` holds the ideas you killed and why. It costs a minute to write and it is the only thing standing between you and re-proposing, in June, the idea you buried in March.

WHY IT CAN BE OPEN

The plugin holds no opinion about your field. Everything field-specific is a file you write, which is also what makes it publishable.

10 What this is not

- **Not a literature database.** It queries real indexes — OpenAlex, Crossref, Semantic Scholar, arXiv, Unpaywall, Europe PMC — through their official APIs. Coverage is theirs, and it is uneven by field. The tool's contribution is refusing to let you *not* look, not pretending the index is complete.
- **Not a substitute for reading.** The read-depth cap exists precisely because the tool cannot read for you. It can only refuse to let an abstract masquerade as a method section.
- **Not a calibrated judge.** The severity classes rank; they do not measure. Where a number was chosen rather than measured, the tool says so and does not block on it.
- **Not a way around a paywall.** Official APIs and legally open full text only.
- **Not an author.** It has no opinion about what you should study, and the parts that look like opinions are your own `field.md` read back to you.

STATED PLAINLY

A research tool that overstates itself would be an unusually poor joke.

11 Where this is

The predecessor of this design is a working harness for software work, in daily use, where the compiler does the blocking. Everything here is the attempt to answer one question honestly — *what plays that part when the work is research* — and then to build only what that answer supports.

The build order follows the confidence. The prose spine and the ledger first, because they change behaviour immediately and cost nothing to revise. The blocking gates second, because that is where hallucinated citations stop being possible. The literature spine third, because it is the largest piece of genuine capability and the one worth doing slowly. The calibration engine last, because it is the part most likely to be wrong on first contact with a real draft.

HONEST STATUS

Design settled, build starting. This document exists so the argument can be attacked before the code makes it expensive to change.

If a claim in this document is wrong, that is a finding and it should be filed as one. The tool this describes would refuse to let it stand without a locator, and the document should be held to the same bar.